

LINGYU(Alvin) XU

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Education

Worcester Polytechnic Institute, Worcester, MA

Jan 2024 - Dec 2025

Master of Science in Information Technology in System Design & Artificial Intelligence

- GPA: 3.81
- **Coursework:** LLM, Application Development, Machine Learning, Data Mining, Data Visualization, SQL

Pennsylvania State University, State College, PA

Aug 2015 - May 2020

Bachelor of Science in Computer Science

- **Coursework:** System Programming, Network Programming, Algorithm, Data Structure, Statistics

Experience

Research Assistant

Worcester, MA

Worcester Polytechnic Institute

June 2025 – Dec 2025

- **LLM Research:** Conducting research on applying large language models to git repo, writing and testing prompt-based code to extract insights, generate summaries, and evaluate model responses. Focused on improving accuracy and relevance of outputs using LLM APIs and Python.

Software Development Engineer I

Shanghai, China

Dell Technologies

Mar 2021 – Dec 2023

- In Dell-APEX day1 team, eliminated manual configuration overhead by building automated APIs (**Python/Go/Java**) for **Azure HCI** integration. The solution automated the entire setup lifecycle for Windows servers—including domain joining, security hardening, and cluster configuration validation—accelerating deployment timelines by an estimated 20%.
- Streamlined the initial server setup by creating a user-friendly (**Angular & Typescript**) interface for Day 1 operations. The UI—featuring intuitive forms and clear data visualizations—guided operators through pre-configuration validation, reducing errors and increasing the efficiency of the entire configuration process by 15%.
- Led a team of 4 in designing and implementing a **Jenkins** CI/CD pipeline for automated sanity testing of new dev images. Managed the entire project lifecycle using **Jira** for task tracking and Git for version control, establishing a robust process that slashed manual effort and boosted code quality by 30%.
- Automated Day-1 infrastructure provisioning by developing containerized microservices (**Docker/Kubernetes**) for storage configuration and admin system initialization, reducing service start failure rate by 25%.

Software Development Engineer Internship

Shanghai, China

Dell Technologies

Mar 2021 – May 2021

- Engineered **Python/Java** APIs to automate the Day-1 setup for Dell VxRail HCI systems, focusing on VMware configuration pre-validation, VxM network setup, and **PostgreSQL** table management. This automation initiative reduced incompatible configuration errors during deployment by 15%.
- Engineered responsive web interfaces with **Angular** for the VxRail Day 1 & Day 2 operations team. Developed UI components (dynamic forms, data filters) to pre-config server configuration for VLAN, authentication and backup settings, reducing manual setup steps and increasing team configuration efficiency by 10%.

Projects

Blockchain Document & Payment Platform, Harvard University Hackathon, Boston, MA

- Developed a full-stack blockchain platform using **Vue.js** and **Node.js** that enables decentralized medical record sharing and **XRP** payment processing, featuring an intuitive front-end for document management and a backend handling business logic and blockchain integration, which reduced sharing evaluation time by 50% and improved insurance tracking through secure smart contracts on **Flare Coston2 testnet**.
- Awarded 3rd Place at the Harvard University Hackathon (EasyA x Flare Network x XRPL Commons) for technical execution and innovation in decentralized healthcare data management.

Android App Development, Worcester Polytechnic Institute, Worcester, MA

- Developed an **Android** application in **Java** and **JavaScript** simulating core banking operations, including balance inquiry, transaction history filtering, and dynamic sorting, backed by a structured **SQLite** database for persistent local storage.
- Engineered a normalized relational schema with interconnected tables; automated data generation using **Python** scripts and implemented backend data handling and query logic in Java to optimize retrieval performance and application responsiveness.

Skills

Languages: Python, Go, Java, Angular, JavaScript, TypeScript, HTML5, CSS, C, C++

DevOps & CI/CD: Docker, Kubernetes, Jenkins, PACT, Git

Tools & Project Management: Azure, REST APIs, Agile, JIRA, Confluence, Mendix Development

Databases & Data Analysis: MySQL, PostgreSQL, MongoDB, SQLite, Tableau, Excel

Data Science & Machine Learning: TensorFlow, PyTorch, Scikit-Learn, Pandas, Spark, Scala, PIG